ALLEN CAREER INSTITUTE
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

p Block Element

English Medium

QUESTION BANK ON p-BLOCK ELEMENTS

EXERCISE # I

Only one option is correct :

1. PH_3(Phosphine) when passed in aqueous solution of $CuSO_4$ it produce -

(A) Blue precipitate of $Cu(OH)_2$ (B) dark blue solution of $[Cu(PH_3)_4]SO_4$

(C) Black precipitate of Cu_3P_2 (D) Colorless solution of $[Cu(H_2O)_4]^+$

PB0001

2. $H_3PO_2 \xrightarrow{\Delta} (X) + PH_3$; is

(A) Dehydration reaction (B) Oxidation reaction

(C) Disproportionation reaction (D) Dephosphorelation reaction

PB0002

3. Which of the following species is not a pseudohalide?

(A) CNO^- (B) $RCOO^-$ (C) OCN^- (D) N_3^-

PB0003

4. An orange solid (X) on heating, gives a colourless gas (Y) and a only green residue (Z). Gas (Y) on treatment with Mg, produces a white solid substance .....

(A) Mg_3N_2 (B) MgO (C) Mg_2O_3 (D) $MgCl_2$

PB0004

5. Conc. HNO_3 is yellow coloured liquid due to

(A) dissolution of NO in conc. HNO_3 (B) dissolution of NO_2 in conc. HNO_3

(C) dissolution of N_2O in conc. HNO_3 (D) dissolution of N_2O_3 in conc. HNO_3

PB0005

6. A gas at low temperature does not react with the most of compounds. It is almost inert and is used to create inert atmosphere in bulbs. The combustion of this gas is exceptionally an endothermic reaction. Based on the given information, we can conclude that the gas is

(A) oxygen (B) nitrogen

(C) carbon mono-oxide (D) hydrogen

PB0006

7. When chlorine gas is passed through an aqueous solution of a potassium halide in the presence of chloroform, a voilet colouration is obtained. On passing more of chlorine water, the voilet colour is disappeared and solution becomes colourless. This test confirms the presence of ..... in aqueous solution.

(A) chlorine (B) fluorine (C) bromine (D) iodine

PB0007

8. $H_3PO_2 \xrightarrow{140°C} A \xrightarrow{220°C} B \xrightarrow{320°C} C$

Compound (C) is

(A) H_2PO_3 (B) H_3PO_3 (C) $(HPO_3)_n$ (D) $H_4P_2O_7$

PB0008

9. An explosive compound (A) reacts with water to produce NH_4OH and HOCl. Then, the compound (A), is

(A) TNG (B) NCl_3 (C) PCl_3 (D) HNO_3

PB0009

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10. An inorganic compound (A) made of two most occuring elements into the earth crust, having a polymeric tetrahedral network structure. With carbon, compound (A) produces a poisonous gas (B) which is the most stable diatomic molecule. Compounds (A) and (B) will be

(A) SiO_2, CO_2 (B) SiO_2, CO (C) SiC,CO (D) SiO_2, N_2

PB0010

11. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S–S bond. Compound (D) with HCI, forms a Lewis acid (E) which exists as a dimer. Compounds (A), (B), (C), (D) and (E) are respectively

(A) $FeSO_4$, SO_2, SO_3, Fe_2O_3, $FeCl_3$ (B) $Al_2(SO_4)_3$, SO_2, SO_3, Al_2O_3, $FeCl_3$

(C) FeS, SO_2, SO_3, $FeSO_4$, $FeCl_3$ (D) FeS, SO_2, SO_3 , $Fe_2(PO_4)_3$, $FeCl_2$

PB0011

12. A tetra-atomic molecule (A) on reaction with nitrogen(I)oxide, produces two substances (B) and (C). (B) is a dehydrating agent while substance (C) is a diatomic gas which shows almost inert behaviour. The substances (A) and (B) and (C) respectively will be

(A) P_4, P_4O_{10}, N_2 (B) P_4, N_2O_5, N_2 (C) P_4, P_2O_3, Ar (D) P_4, P_2O_3, H_2

PB0012

13. First compound of inert gases was prepared by scientist Neil Bartlett in 1962. This compound is

(A) $XePtF_6$ (B) XeO_3 (C) XeF_6 (D) $XeOF_4$

PB0013

14. Carbongene has X% of CO_2 and is used as an antidote for poisoning of Y . Then, X and Y are

(A) X = 95% and Y = lead poisoning (B) X = 5% and Y = CO poisoning

(C) X = 30% and Y = CO_2 poisoning (D) X = 45% and Y = CO poisoning

PB0014

15. The correct order of acidic strength of oxides of nitrogen is

(A) $NO < NO_2 < N_2O < N_2O_3 < N_2O_5$ (B) $N_2O < NO < N_2O_3 < N_2O_4 < N_2O_5$

(C) $NO < N_2O < N_2O_3 < N_2O_5 < N_2O_4$ (D) $NO < N_2O < N_2O_5 < N_2O_3 < N_2O_4$

PB0015

16. $H_3BO_3 \xrightarrow{T_1} X \xrightarrow{T_2} Y \xrightarrow{\text{red hot}} B_2O_3$

if $T_1 < T_2$ then X and Y respectively are

(A) X = Metaboric acid and Y = Tetraboric acid

(B) X = Tetraboric acid and Y = Metaboric acid

(C) X = Borax and Y = Metaboric acid

(D) X = Tetraboric acid and Y = Borax

PB0016

17. When conc. H_2SO_4 was treated with $K_4[Fe(CN)_6]$, CO gas was evolved. By mistake, somebody used dilute H_2SO_4 instead of conc. H_2SO_4 then the gas evolved was

(A) CO (B) HCN (C) N_2 (D) CO_2

PB0017

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18. An inorganic white crystalline compound (A) has a rock salt structure. (A) on reaction with conc. H_2SO_4 and MnO_2, evolves a pungent smelling, greenish-yellow gas (B). Compound (A) gives white ppt. of (C) with $AgNO_3$ solution. Compounds (A), (B) and (C) will be respectively

(A) NaCl, Cl_2, AgCl (B) NaBr, Br_2, NaBr

(C) NaCl, Cl_2, Ag_2SO_4 (D) Na_2CO_3 , CO_2, Ag_2CO_3

PB0018

19. $RCl \xrightarrow[Si]{Cu-powder} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{condensation} A$

Compound (A) is

(A) a linear silicone (B) a chlorosilane (C) a linear silane (D) a network silane

PB0019

20. When oxalic acid reacts with conc. H_2SO_4 , two gases produced are of neutral and acidic in nature respectively. Potassium hydroxide absorbs one of the two gases. The product formed during this absorption and the gas which gets absorbed are respectively

(A) K_2CO_3 and CO_2 (B) $KHCO_3$ and CO_2

(C) K_2CO_3 and CO (D) $KHCO_3$ and CO

PB0020

21. Conc. H_2SO_4 cannot be used to prepare HBr from NaBr because it

(A) reacts slowly with NaBr (B) oxidises HBr

(C) reduces HBr (D) disproportionates HBr

PB0021

22. Ammonia can be dried by

(A) conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) anhydrous $CaCl_2$

PB0022

23. When chlorine reacts with a gas X, an explosive inorganic compound Y is formed. Then X and Y will be

(A) X = O_2 and Y = NCl_3 (B) X = NH_3 and Y=NCl_3

(C) X = O_2 and Y = NH_4Cl (D) X = NH_3 and Y = NH_4Cl

PB0023

24. $HNO_3 + P_4O_{10} \longrightarrow HPO_3 + A$; the product A is

(A) N_2O (B) N_2O_3 (C) NO_2 (D) N_2O_5

PB0024

25. Which of the following is the correct order of acidic strength?

(A) $Cl_2O_7 > SO_3 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

PB0025

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26. $Ca + C_2 \longrightarrow CaC_2 \xrightarrow{N_2} A$

Compound (A) is used as a/an

(A) fertilizer (B) dehydrating agent (C) oxidising agent (D) reducing agent

PB0026

27. A gas which exists in three allotropic forms α, β and γ is

(A) SO_2 (B) SO_3 (C) CO_2 (D) NH_3

PB0027

28. A red coloured mixed oxide (X) on treatment with conc. HNO_3 gives a compound (Y). (Y) with HCl, produces a chloride compound (Z) which can also be produced by treating (X) with conc. HCl. Compounds (X) , (Y), and (Z) will be

(A) Mn_3O_4, MnO_2, $MnCl_2$ (B) Pb_3O_4, PbO_2, $PbCl_2$

(C) Fe_3O_4, Fe_2O_3, $FeCl_2$ (D) Fe_3O_4, Fe_2O_3, $FeCl_3$

PB0028

29. One mole of calcium phosphide on reaction with excess of water gives

(A) one mole of phosphine (B) two moles of phosphoric acid

(C) two moles of phosphine (D) one mole of phosphorus penta-oxide

PB0029

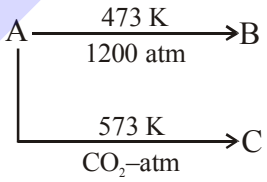
30. $NaH_2PO_4 \xrightarrow{> 240°C} \underset{\text{Sodium trimetaphosphate}}{(NaPO_3)_3} \xrightarrow{625°C} NaPO_3(\text{liquid melt}) \xrightarrow[\text{cooling}]{\text{rapid}} D(\text{glass})$

Compound (D) is known as

(A) Microcosmic salt (B) Graham's salt (C) Fischer's salt (D) Switzer's Salt

PB0030

31. Three allotropes (A), (B) and (C) of phosphorous in the following change are respectively



(A) white, β-black, red (B) β-black, white, red

(C) red, β-black, white (D) red, violet, β-black

PB0031

32. When an inorganic compound reacts with SO_2 in aqueous medium, produces (A). (A) on reaction with Na_2CO_3, gives compound (B) which with sulphur, gives a substance (C) used in photography. Compound (C) is

(A) Na_2S (B) $Na_2S_2O_7$ (C) Na_2SO_4 (D) $Na_2S_2O_3$

PB0032

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33. $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$

How can this reaction is made to proceed in forward direction?

(A) addition of cis 1,2 diol (B) addititon of borax

(C) addition of trans 1,2 diol (D) addition of Na_2HPO_4

PB0033

34. Which is the compound responsible for the flickering light called **will-o-the-wisp**, some times seen in the Marsh.

(A) PH_3 (B) P_2H_4 (C) H_2S (D) $PH_3 + H_2S$

PB0034

35. The gun powder is consisting of'_____'+ sulphur + Charcoal what is the missing substance for gun powder

(A) $LiNO_3$ (B) NH_4NO_2 (C) KNO_3 (D) (A) and (B) mixture

PB0035

36. An aqueous solution of borax is

(A) Neutral (B) Amphoteric (C) Basic (D) Acidic

PB0036

37. Boric acid is polymeric due to

(A) Its acidic nature (B) The presence of hydrogen bonds

(C) Its monobasic nature (D) Its geometry

PB0037

38. The type of hybridisation of boron in diborane is

(A) sp (B) sp^2 (C) sp^3 (D) dsp^2

PB0038

39. Thermodynamically the most stable form of carbon is

(A) Diamond (B) Graphite (C) Fullerenes (D) Coal

PB0039

40. Elements of group 14

(A) Exhibit oxidation state of +4 only (B) Exhibit oxidation state of +2 and +4 only

(C) Form M^{2-} and M^{4+} ions (D) Form M^{2+} and M^{4+} ions

PB0040

41. $A + Br_2 \rightarrow N_2 + (B)$

$\downarrow$ NaOH

(A)

if A is a basic gas then identified (A) and (B)

(A) NH_3, NH_4Br (B) NH_3, N_2O (C) NH_3, N_2O_5 (D) None of these

PB0041

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Question No. 42 to 47 (6 questions)

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

42. **Assertion :** Borax bead test is applicable only to coloured salt.

Reason : In borax bead test, coloured salts are decomposed to give coloured metal meta borates.

PB0042

43. **Assertion :** Aluminium and zinc metal evolve H_2 gas from NaOH solution

Reason : Several non-metals such as P, S, Cl, etc. convert a hydride instead of producing H_2 gas from NaOH.

PB0043

44. **Assertion :** Conc. H_2SO_4 can not be used to prepare pure HBr from NaBr

Reason : It reacts slowly with NaBr.

PB0044

45. **Assertion :** Oxygen is more electronegative than sulphur, yet H_2S is acidic, while H_2O is neutral.

Reason : H–S bond is weaker than O–H bond.

PB0045

46. **Assertion :** Chlorine gas disproportionates in hot & conc. NaOH solution.

Reason : NaCl and NaOCl are formed in the above reaction.

PB0046

47. **Assertion :** Liquid IF_5 conducts electricity.

Reason : Liquid IF_5 self ionizes as, $2IF_5 \rightleftharpoons IF_4^+ + IF_6^-$

PB0047

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EXERCISE # II

***One or more than one option may be correct* :**

1. When a compound X reacts with ozone in aqueous medium, a compound Y is produced. Ozone also reacts with Y and produces compound Z. Z acts as an oxidising agent, then X, Y and Z will be

(A) X = HI, Y = I_2 and Z = HIO_3

(B) X = KI, Y = I_2 and Z = HIO_3

(C) X = KI, Y = I_2 and Z = HIO_4

(D) X = HI, Y = I_2 and Z = HIO_4

PB0048

2. Which of the following statements is/are correct regarding B_2H_6?

(A) banana bonds are longer but stronger than normal B–H bonds

(B) B_2H_6 is also known as 3c–2e compound

(C) the hybrid state of B in B_2H_6 is sp^3 while that of sp^2 in BH_3

(D) it cannot be prepared by reacting BF_3 with $LiBH_4$ in the presence of dry ether

PB0049

3. Which of the following statements is/are correct regarding inter-halogen compounds of AB_x type?

(A) x may be 1,3,5 and 7

(B) A is a more electronegative halogen than B

(C) FBr_3 cannot exit

(D) The interhalogens are generally more reactive than the halogens (except F_2) due to weaker A–X bonds compared to X–X bond.

PB0050

4. When an inorganic compound (X) having 3c-2e as well as 2c-2e bonds reacts with ammonia gas at a certain temperature, gives a compound (Y) iso-structural with benzene. Compound (X) with ammonia at a high temperature, produces a slippery substance (Z). Then

(A) (X) is B_2H_6

(B) (Z) is known as inorganic graphite

(C) (Z) having structure similar to graphite

(D) (Z) having structure similar to (X)

PB0051

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5. Boric acid

(A) exists in polymeric form due to inter-molecular hydrogen bonding.

(B) is used in manufacturing of optical glasses.

(C) is a tri-basic acid

(D) with borax, it is used in the preparation of a buffer solution.

PB0052

6. The correct statement(s) related to allotropes of carbon is/are

(A) graphite is the thermodynamically most stable allotrope of carbon and having a two dimensional sheet like structure of hexagonal rings of carbon (sp^2)

(B) diamond is the hardest allotrope of carbon and having a three dimensional network structure of C(sp^3)

(C) fullerene (C_{60}) is recently discovered non-crystalline allotrope of carbon having a football-like structure.

(D) Vander Waal's force of attraction acts between the layers of graphite 6.14 Å away from each other

PB0053

7. $Al_2(SO_4)_3 + NH_4OH \longrightarrow X$, then

(A) X is a white coloured compound

(B) X is insoluble in excess of NH_4OH

(C) X is soluble in NaOH

(D) X cannot be used as an antacid

PB0054

8. The species that undergo(es) disproportionation in an alkaline medium is/are

(A) Cl_2 (B) MnO_4^{2-} (C) P_4 (D) ClO_4^-

PB0055

9. Select correct statement(s):

(A) Borax is used as a buffer

(B) 1 M borax solution reacts with equal volumes of 2 M HCl solution

(C) Titration of borax can be made using methyl orange as the indicator

(D) Coloured bead obtained in borax-bead test contains metaborate

PB0056

10. Which of the following is / are correct for group 14 elements?

(A) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$

(B) The ability to form $p\pi-p\pi$ multiple bonds among themselves increases down the group

(C) The tendency for catenation decreases down the group

(D) They all form oxides with the formula MO_2.

PB0057

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11. Zeolite is used in which of the following cases :

(A) Conversion of alcohols into gasoline (B) Cracking of hydrocarbon

(C) Isomerisation of hydrocarbons (D) Softening of hard water

PB0058

12. Which of the following oxides are mixed oxide ?

(A) PbO_2 (B) SnO_2 (C) Pb_2O_3 (D) Pb_3O_4

PB0059

13. Which of the following oxide(s) gives brown ppt on reaction with conc. HNO_3 :

(A) PbO (B) SnO (C) Pb_2O_3 (D) Pb_3O_4

PB0060

14. Which of the following reaction produces PH_3 :

(A) $Ca_3P_2 + H_2O \rightarrow$ (B) $P_4 + NaOH \rightarrow$ (C) $PH_4I + KOH \rightarrow$ (D) $H_3PO_2 \xrightarrow{\Delta}$

PB0061

15. Which of the following halides is least stable and has doubtful existence ?

(A) CCl_4 (B) GeI_4 (C) SnI_4 (D) PbI_4

PB0062

16. $Ca_2B_6O_{11} + Na_2CO_3 \xrightarrow{\Delta} [X] + CaCO_3 + NaBO_2$ (Unbalanced equation)

Correct statement for [X]

(A) Structure of anion of crystalline (X) has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised

(B) (X) with NaOH(aq.) gives a compound which on reaction with H_2O_2 in alkaline medium yields a compound used as brightner in soaps

(C) Hydrolysis of (X) with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid

(D) [X] on heating with cobalt salt in oxidising flame gives blue coloured bead

PB0063

17. (A) + 2C $\xrightarrow[2500°C]{2000°C \text{ to}}$ (B) +2CO ↑

(B) + Carbon $\xrightarrow[2500°C]{2000°C \text{ to}}$ (C)

If A is an example of 3-d silicate then select the correct statements about (C)

(A) Central atom of C is sp^3 hybridised

(B) (C) is non planar and all atoms are sp^3 hybridised

(C) (C) has diamond like structure, and it is coloured when impurity is present but pale yellow to colourless solid at room temperature

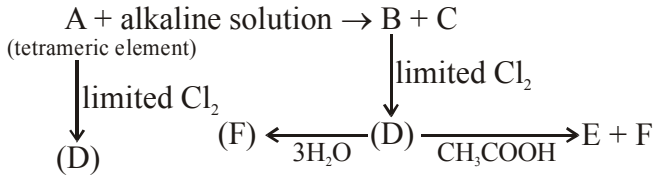
(D) (C) is silicon carbide (SiC) and it is not being affected by any acid except H_3PO_4

PB0064

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EXERCISE # III

Paragraph for Question No. 1 & 2



1. When D react with C_2H_5OH then product will be

(A) C_2H_5Cl, H_3PO_4 (B) C_2H_5Cl, H_3PO_3

(C) CH_3COCl, H_3PO_3 (D) Only H_3PO_3

PB0065

2. B can be absorbed by :

(A) Ca(OCl) Cl (B) H_2S (C) Both (D) None

PB0065

Paragraph for Question No. 3 to 6

$$HCOOH \xrightarrow[Conc.H_2SO_4]{373K} H_2O + (X)$$

$$C(s) + H_2O \xrightarrow{423-1273K} (X) + H_2(g)$$

3. Select the correct statement about (X)

(A) (X) is a colourless, odourless and almost water insoluble gas

(B) X is highly poisonous and burns with blue flame

(C) When (X) gas is passed through $PdCl_2$ solution giving rise to black ppt

(D) All of these

PB0066

4. Mixture of (X) gas + H_2 is called

(A) Water gas or synthesis gas (B) Producer gas

(C) Methane gas (D) None of these

PB0066

5. In second reaction when air is used instead of steam a mixture of (X) gas and N_2 is produced which is called

(A) Water gas (B) Synthesis gas (C) Producer gas (D) Carbon dioxide gas

PB0066

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6. Select the correct statement about (X)

(A) (X) gas is estimated by I_2O_5 (B) Cu_2Cl_2 is absorber of (X) gas

(C) (X) gas is the purifying agent for Ni (D) All of these

PB0066

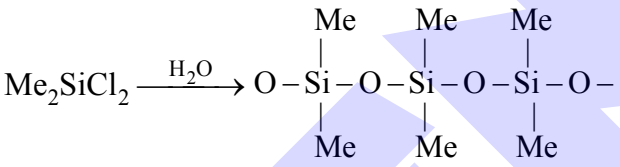
Paragraph for Question No. 7 & 8

Read the following write-ups and answer the questions at the end of it.

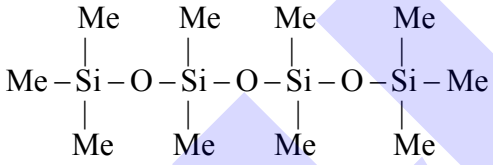
Silicons are synthetic polymers containing repreated R_2SiO units. Since, the empirical formula is that of a ketone (R_2CO), the name silicone has been given to these materials. Silicones can be made into oils, rubbery elastomers and resins. They find a variety of applications because of their chemical inertness, water repelling nature, heat-resistance and good electrical insulating property.

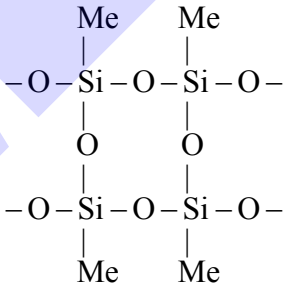
Commercial silicon polymers are usually methyl derivatives and to a lesser extent phenyl derivatives and are synthesised by the hydrolysis of

R_2SiCl_2[R=methyl (Me) or phenyl (ϕ)]



7. If we mix Me_3SiCl with Me_2SiCl_2, we get silicones of the type:

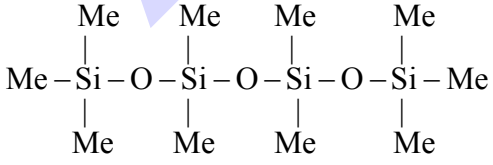
(A) 

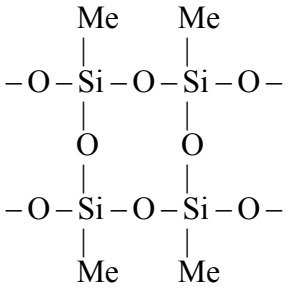
(B) 

(C) both of the above (D) none of the above

PB0067

8. If we start with $MeSiCl_3$ as the starting material, silicones formed is:

(A) 

(B) 

(C) Both of the above (D) None of the above

PB0067

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Paragraph for Question No. 9 & 10

CO gas is absorbed by aqueous suspension of cuprous chloride forming the complex like $[CuCl(CO)(H_2O)_2]$.

9. Comment on the shape of the above complex.

(A) Tetrahedral (B) TBP (C) Square planar (D) Cannot be predicted

PB0068

10. Choose the correct statement regarding the above molecule

(A) Cl-atom is separated by equal angle from both of the water molecule

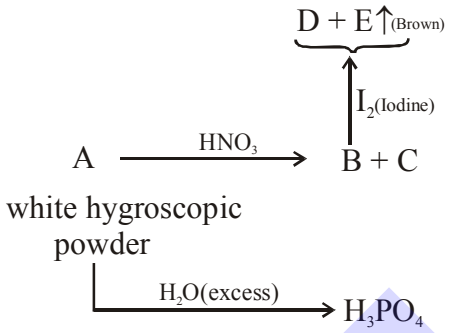
(B) Magnetic moment of the above complex is 1.73 B.M.

(C) There are two stereo isomer for the above complex.

(D) Both (A) and (C)

PB0068

Paragraph for Question No. 11 to 12



11. Which of the following compound is not having +5 oxidation state on its central atom

(A) B (B) D (C) C (D) E

PB0069

12. "A" on reacting with Acetamide yields a compound "F". The compound "F" contains

(A) 4 σ bonds, 2 π bonds

(B) 5 σ bonds, 1 π bonds

(C) 5 σ bonds, 2 π bonds

(D) 2 σ bonds, 2 π bonds

PB0069

13. Match List-I with List-II

List-I (Chemical reaction)	**List-II (Name of process)**
(I) $4NH_3 + 5O_2 \xrightarrow{800°C/Pt} 4NO + 6H_2O$	(a) Contact process
(II) $4HCl + O_2 \xrightarrow{230°C/CuCl_2} 2Cl_2 + 2H_2O$	(b) Ostwald's process
(III) $2SO_2 + O_2 \xrightarrow{450-500°/V_2O_5} 2SO_3$	(c) Deacon's process
(IV) $2N_2 + 3H_2 \xrightarrow{Fe+Mo} 2NH_3$	(d) Haber's proces

(A) I-a, II-b, III-d, IV-c (B) I-b, II-c, III-a, IV-d

(C) I-a, II-d, III-c, IV-b (D) I-a, II-c, III-b, IV-d

PB0070

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14. **Column-I** **Column-II**

(P) Dry ice (1) Used as antidote for CO-poisoning

(Q) Carbongene (2) Used as nonstick coating

(R) Carborundum (3) Used as refrigerant

(S) Teflon (4) Used as abrassive

Code :

	P	Q	R	S
(A)	4	1	3	2
(C)	3	1	4	2

	P	Q	R	S
(B)	4	2	1	3
(D)	1	4	3	2

PB0071

15. **Column-I** **Column-II**

Compound **Correct statement for compounds given**

(P) $SnCl_2$ (1) Used in printing technology

(Q) Butter of tin (2) Used for gilding purpose (in joining gold pieces)

(R) Mosaic gold (3) Reducing agent

(S) Pink's salt (4) Mordant

Code :

	P	Q	R	S
(A)	4	3	2	1
(C)	2	1	3	4

	P	Q	R	S
(B)	3	4	2	1
(D)	1	3	4	2

PB0072

16. **Column-I (Metal)** **Column-II (Correct statements)**

(P) Fe (1) Produces NO with 20% HNO_3

(Q) Cu (2) Produces NH_4NO_3 with 6% HNO_3

(R) Pb (3) Produces NO_2 with 70% HNO_3

(S) Sn (4) Produces NH_4NO_3 with 20% HNO_3

Code :

	P	Q	R	S
(A)	2, 1	1	3, 4	2, 3
(B)	2, 3	1, 3	1, 3	2, 3, 4
(C)	1, 3	1, 2	3, 4	2
(D)	1, 4	2, 3	1, 3	1, 4

PB0073

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17. **Column-I (Reactions)** **Column-II (Correct statements)**

(P) $XeF_2 + PF_5 \rightarrow$ (1) Fluoride of Xe acts as fluoride acceptor

(Q) $XeF_4 + Pt \rightarrow$ (2) Fluoride of Xe undergoes disproportion

(R) $XeF_4 + H_2O \rightarrow$ (3) Fluoride of Xe acts as fluorinating agent

(S) $XeF_6 + CsF \rightarrow$ (4) Fluoride of Xe act as fluoride donor

Code :

	P	Q	R	S
(A)	4	2	3	1
(C)	4	3	2	1

	P	Q	R	S
(B)	3	2	1	4
(D)	3	4	2	1

PB0074

18. **Column-I (Substances)** **Column-II (Can be prepared by)**

(P) O_3 (1) Acidification of BaO_2 with H_3PO_4

(Q) Bleaching powder (2) Birkeland Eyde process

(R) H_2O_2 (3) Dry O_2 is passed through a silent electrical discharge

(S) HNO_3 (4) Cl_2 gas is passed through slaked lime

Code :

	P	Q	R	S
(A)	3	4	1	2
(C)	2	1	3	4

	P	Q	R	S
(B)	1	3	4	2
(D)	4	1	2	3

PB0075

MATCHING LIST TYPE 1 × 3 Q. (19 to 21)

For the following molecules in column-I, match the correct order of properties with column-II & column-III according to the questions asked.

Column - 1 Molecules	Column - 2 Properties	Column - 3 Correct order
(I) NH_3, PH_3, AsH_3, SbH_3	(i) Bond angle	(p) Increasing order
(II) H_2O, H_2S, H_2Se, H_2Te	(ii) Reducing character	(q)Decreasing order
(III) HF, HCl, HBr, HI	(iii) Intermolecular force	(r) All equal
(IV) CH_4, SiH_4, GeH_4, SnH_4	(iv) Boiling Point	(s) Irregular order

19. Which is the only **CORRECT** combination?

(A) (I), (iv), (p) (B) (II), (i), (p) (C) (III), (iv), (q) (D) (IV), (iii), (p)

PB0076

20. Which is the only **INCORRECT** combination?

(A) (I), (ii), (p) (B) (IV), (i), (r) (C) (III), (iii), (q) (D) (II), (iv), (s)

PB0076

21. In which combination, Drago's Rule plays an insignificant role in prediction of orders?

(A) (I), (i), (q) (B) (II), (iv), (s) (C) (IV), (i), (r) (D) (III), (ii), (p)

PB0076

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MATCHING LIST TYPE 1 × 3 Q. (22 to 24)

Molecule	Hybridization of central atom(s)	Properties related to structure
(P) $(HBNH)_3$	(I) sp^2	(i) Cyclic structure
(Q) C_3O_2	(II) sp^3	(ii) Has (X–$\ddot{O}$–X) linkage (X = central atom)
(R) $P_4O_{12}^{4-}$	(III) sp^3d	(iii) Planar
(S) N_2O_4	(IV) sp	(iv) Has (X – X) linkage

22. Identify **CORRECT** match

(A) P, II, iii (B) P, I, i (C) P, IV, iii (D) Q, I, i

PB0077

23. Identify **CORRECT** match

(A) Q, IV, iii (B) Q, IV, i (C) Q, I, ii (D) R, I, ii

PB0077

24. Identify **CORRECT** match

(A) R, II, iv (B) R, II, ii (C) S, III, ii (D) S, IV, i

PB0077

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EXERCISE # JEE-MAINS

1. Which products are expected from the disproportionation of hypochlorous acid : **[AIEEE-2002]**

(1) $HClO_3$ and Cl_2O (2) $HClO_2$ and HClO (3) HCl and Cl_2O (4) HCl and $HClO_3$

PB0078

2. Identify the **INCORRECT** statement among the following : **[AIEEE-2002]**

(1) Ozone reacts with SO_2 to give SO_3

(2) Silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3 and H_2O

(3) Cl_2 reacts with excess of NH_3 to give N_2 and HCl

(4) Br_2 reacts with hot and strong NaOH solution to give NaBr, $NaBrO_4$ and H_2O

PB0079

3. Aluminium is industrially prepared by: **[AIEEE-2002]**

(1) Fused cryolite (2) Bauxite ore (3) Alunite (4) Borax

PB0080

4. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be : **[AIEEE-2003]**

(1) Sodium (2) Magnesium (3) Mercury (4) Tin

PB0081

5. What may be expected when phosphine gas is mixed with chlorine gas: **[AIEEE-2003]**

(1) PCl_5 and HCl are formed and mixture cools down

(2) $PH_3.Cl_2$ is formed with warming up

(3) The mixture only cools down

(4) PCl_3 and HCl are formed and the mixture warms up

PB0082

6. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite : **[AIEEE-2003]**

(1) Has molecules of variable molecular masses like polymers

(2) Has carbon atoms arranged in large plated of rings of strongly bonded carbon atoms with weak interplate bonds

(3) Is a non crystalline substance

(4) Is an allotropic form of diamond

PB0083

7. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. This is due to : **[AIEEE-2003]**

(1) Strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke

(2) Due to strong affinity for water, conc. HCl pulls moisture of air towards self. The moisture forms droplets of water and hence the cloud

(3) Conc. HCl emits strongly smelling HCl gas all the time

(4) Oxygen in air reacts with emitted HCl gas to form a cloud of Cl_2 gas

PB0084

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8. Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives- **[AIEEE-2004]**

(1) $Al^{3+} + 3Cl^-$

(2) $[Al(H_2O)_6]^{3+} + 3Cl^-$

(3) $[Al(OH)_6]^{3-} + 3HCl$

(4) $Al_2O_3 + 6HCl$

PB0085

9. The soldiers of Napolean army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to:- **[AIEEE-2004]**

(1) An interaction with water vapour contained in humid air

(2) A change in crystalline structure of tin

(3) A change in the partial pressure of O_2 in air

(4) An interaction with N_2 of air at low temperature

PB0086

10. Which one of the following statements regarding helium is incorrect **[AIEEE-2004]**

(1) It is used to produce and sustain powerful superconducting magnets

(2) It is used as a cryogenic agent for carrying out experiments at low temperatures

(3) It is used to fill gas balloons instead of hydrogen because it is lighter than hydrogen and non-inflammable

(4) It is used in gas-cooled nuclear reactors

PB0087

11. The number of hydrogen atoms attached to phosphorus atom in hypophosphorous acid is : **[AIEEE-2005]**

(1) Zero (2) Two (3) One (4) Three

PB0088

12. Heating an aqueous solution of aluminium chloride to dryness will give :- **[AIEEE-2005]**

(1) $AlCl_3$ (2) Al_2Cl_6 (3) Al_2O_3 (4) $Al(OH)Cl_2$

PB0089

13. Which one of the following is the correct statement **[AIEEE-2005]**

(1) Boric acid is a protonic acid

(2) Beryllium exhibits coordination number of six

(3) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase

(4) B_2H_6, $2NH_3$ is known as "inorganic benzene"

PB0090

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14. In silicon dioxide : **[AIEEE-2005]**

(1) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms

(2) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms

(3) Silicon atom is bonded to two oxygen atoms

(4) There are double bonds between silicon and oxygen atoms

PB0091

15. Regular use of which of the following fertilizer increases the acidity of soil : **[AIEEE-2007]**

(1) Potassium nitrate (2) Urea

(3) Superphosphate of lime (4) Ammonium sulphate

PB0092

16. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence: **[AIEEE-2007]**

(1) $GeX_2 << SiX_2 << SnX_2 << PbX_2$ (2) $SiX_2 << GeX_2 << PbX_2 << SnX_2$

(3) $SiX_2 << GeX_2 << SnX_2 << PbX_2$ (4) $PbX_2 << SnX_2 << GeX_2 << SiX_2$

PB0093

17. Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is **[AIEEE-2008]**

(1) R_4Si (2) $RSiCl_3$ (3) R_2SiCl_2 (4) R_3SiCl

PB0094

18. Which one of the following reactions of Xenon compounds is not feasible ? **[AIEEE-2009]**

(1) $2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$ (2) $XeF_6 + RbF \rightarrow Rb[XeF_7]$

(3) $XeO_3 + 6HF \rightarrow XeF_6 + 3H_2O$ (4) $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + 1.5O_2$

PB0095

19. Which of the following statement is wrong ? **[AIEEE-2011]**

(1) Single N–N bond is weaker than the single P–P bond

(2) N_2O_4 has two resonance structures

(3) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table

(4) Nitrogen cannot form $d\pi$-$p\pi$ bond

PB0096

20. Which of the following statements regarding sulphur is incorrect ? **[AIEEE-2011]**

(1) At 600°C the gas mainly consists of S_2 molceules

(2) The oxidation state of sulphur is never less than +4 in its compounds

(3) S_2 molecule is paramagnetic

(4) The vapour at 200°C consists mostly of S_8 rings

PB0097

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21. Boron cannot form which one of the following anions ? **[AIEEE-2011]**

(1) $B(OH)_4^-$ (2) BO_2^- (3) BF_6^{3-} (4) BH_4^-

PB0098

22. In view of the signs of $\Delta_r G°$ for the following reactions

$PbO_2 + Pb \rightarrow 2\ PbO$, $\Delta_r G° < 0$

$SnO_2 + Sn \rightarrow 2\ SnO$, $\Delta_r G° > 0$,

Which oxidation states are more characteristic for lead and tin ? **[AIEEE-2011]**

(1) For lead + 4, for tin + 2 (2) For lead + 2, for tin + 2

(3) For lead + 4, for tin + 4 (4) For lead + 2, for tin + 4

PB0099

23. The number of S–S bonds in SO_3, $S_2O_3^{2-}$, $S_2O_6^{2-}$ and $S_2O_8^{2-}$ respectively are :- **[Jee Main(Online)-2012]**

(1) 1, 0, 1, 0 (2) 0, 1, 1, 0 (3) 1, 0, 0, 1 (4) 0, 1, 0, 1

PB0100

24. Which one of the following depletes ozone layer ? **[Jee Main(Online)-2012]**

(1) NO and freons (2) SO_2 (3) CO (4) CO_2

PB0101

25. In which of the following arrangements, the sequence is not strictly according to the property written against it ? **[Jee Main(Online)-2012]**

(1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

(2) B < C < O < N : increasing first ionisation enthalpy

(3) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(4) HF < HCl < HBr < HI : increasing acid strength

PB0102

26. The formation of molecular complex $BF_3 - NH_3$ results in a change in hybridisation of boron :-

(1) from sp^3 to sp^3d (2) from sp^2 to dsp^2 **[JEE(Main) Online–2012]**

(3) from sp^3 to sp^2 (4) from sp^2 to sp^3

PB0103

27. The catenation tendency of C, Si and Ge is in the order Ge < Si < C. The bond energies (in kJ mol^{-1}) of C — C, Si —Si and Ge—Ge bonds are respectively : **[JEE(Main) Online–2013]**

(1) 348, 260, 297 (2) 348, 297, 260 (3) 297, 348, 260 (4) 260, 297, 348

PB0104

28. The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4, on hydrolysis gives a white gelatinous precipitate. The precipitate is: **[Jee Main(Online)-2014]**

(1) silica gel (2) silicic acid

(3) hydrofluosilicic acid (4) calciumfluorosilicate

PB0105

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29. Which of the following series correctly represents relations between the elements from X to Y?

[Jee Main(Online)-2014]

	$X \rightarrow Y$
(1) $_{18}Ar \rightarrow {}_{54}Xe$	Noble character increases
(2) $_{3}Li \rightarrow {}_{19}K$	Ionization enthalpy increases
(3) $_{6}C \rightarrow {}_{32}Ge$	Atomic radii increases
(4) $_{9}F \rightarrow {}_{35}Br$	Electron gain enthalpy with negative sign increases

PB0106

30. Which of the following statements about the depletion of ozone layer is correct?

[Jee Main(Online)-2014]

(1) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by $Cl^{\bullet}$ and $ClO^{\bullet}$ radicals

(2) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $ClO^{\bullet}$ radicals

(3) Oxides of nitrogen also do not react with ozone in stratosphere

(4) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere

PB0107

31. Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides ?

[Jee Main(Online)-2014]

(1) XeO_2F_2 (2) XeO_3 (3) XeO_4 (4) $XeOF_4$

PB0108

32. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium ?

[Jee Main(Online)-2014]

(1) MnO_4^- (2) SO_3^{2-} (3) KI (4) $Cr_2O_7^{2-}$

PB0109

33. Consider the reaction **[Jee Main(Online)-2014]**

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct?

(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation

PB0110

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34. In the following sets of reactants which two sets best exhibit the amphoteric character of $Al_2O_3.\ xH_2O$? **[JEE(Main) Online–2014]**

Set-1 : $Al_2O_3.xH_2O(s)$ and OH^- (aq)

Set-2 : $Al_2O_3.xH_2O(s)$ and H_2O (ℓ)

Set-3 : $Al_2O_3.xH_2O(s)$ and H^+ (aq)

Set-4 : $Al_2O_3.xH_2O(s)$ and NH_3 (aq)

(1) 1 and 2 (2) 2 and 4 (3) 1 and 3 (4) 3 and 4

PB0111

35. Which of the following compounds has a P-P bond :- **[Jee Main(Online)-2015]**

(1) $H_4P_2O_5$ (2) $(HPO_3)_3$ (3) $H_4P_2O_7$ (4) $H_4P_2O_6$

PB0112

36. Chlorine water on standing loses its colour and forms :- **[Jee Main(Online)-2015]**

(1) HCl and $HClO_2$ (2) HCl only

(3) HOCl and $HOCl_2$ (4) HCl and HOCl

PB0113

37. Which among the following is the most reactive ? **[Jee Main-2015]**

(1) I_2 (2) ICl (3) Cl_2 (4) Br_2

PB0114

38. Which one has the highest boiling point ? **[Jee Main-2015]**

(1) Kr (2) Xe (3) He (4) Ne

PB0115

39. From the following statements regarding H_2O_2, choose the incorrect statement : **[Jee Main-2015]**

(1) It has to be stored in plastic or wax lined glass bottles in dark

(2) It has to be kept away from dust

(3) It can act only as an oxidizing agent

(4) It decomposes on exposure to light

PB0116

40. The reaction of zinc with dilute and concentrated nitric acid, respectively produces : **[JEE (Main) 2016]**

(1) NO_2 and N_2O (2) N_2O and NO_2

(3) NO_2 and NO (4) NO and N_2O

PB0117

41. The non-metal that does not exhibit positive oxidation state is : **[JEE (Main) 2016]**

(1) Oxygen (2) Fluorine (3) Iodine (4) Chlorine

PB0118

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42. Which intermolecular force is most responsible in allowing xenon gas to liquefy?

[JEE (Main) Online 2016]

(1) Ionic

(2) Instantaneous dipole- induced dipole

(3) Dipole - dipole

(4) Ion - dipole

PB0119

43. The following statements concern elements in the periodic table. Which of the following is true ?

(1) The group 13 elements are all metals. **[JEE (Main) Online 2016]**

(2) For group 15 elements, the stability of +5 oxidation state increases down the group.

(3) All the elements in Group 17 are gases.

(4) Elements of group 16 have lower ionization enthalpy values compared to those of group 15 in the corresponding periods.

PB0120

44. ***Assertion :*** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity. **[JEE (Main) Online 2016]**

Reason : Hybridization of carbon in diamond and graphite are sp^3 and sp^2, respectively.

(1) Assertion is incorrect statement, but the reason is correct.

(2)Both assertion and reason are correct, and the reason is the correct explanation for the assertion.

(3) Both assertion and reason are incorrect.

(4) Both assertion and reson are correct, but the reason is not the correct explanation for the assertion.

PB0121

45. Identify the incorrect statement : **[JEE (Main) Online 2016]**

(1) S_8 ring has a crown shape.

(2) The S–S–S bond angles in the S_8 and S_6 rings are the same

(3) S_2 is paramagnetic like oxygen

(4) Rhombic and monoclinic sulphur have S_8 molecules.

PB0122

46. Which of the following reactions is an example of a redox reaction ? **[JEE (Main) 2017]**

(1) $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$

(2) $XeF_2 + PF_5 \rightarrow [XeF]^+PF_6^-$

(3) $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

(4) $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$ **PB0123**

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47. The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :-

(1) ClO^- and ClO_3^- (2) ClO_2^- and ClO_3^- **[JEE (Main) 2017]**

(3) Cl^- and ClO^- (4) Cl^- and ClO_2^-

PB0124

48. In which of the following reaction, hydrogen peroxide acts as an oxidizing agent ?

(1) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$ **[JEE (Main) ONLINE 2017]**

(2) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(4) $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$

PB0125

49. XeF_6 on partial hydrolysis with water produces a compound 'X'. The same compound 'X' is formed when XeF_6 reacts with silica. The compound 'X' is:- **[JEE (Main) ONLINE 2017]**

(1) XeO_3 (2) XeF_4 (3) $XeOF_4$ (4) XeF_2

PB0126

50. The compound that does not produce nitrogen gas by the thermal decomposition is

[JEE (Main) OFFLINE 2018] (1)

$(NH_4)_2Cr_2O_7$ (2) NH_4NO_2 (3) $(NH_4)_2SO_4$ (4) $Ba(N_3)_2$

PB0127

51. For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ? **[JEE (Main) ONLINE 2018]**

(Given : Atomic wt. – Cr = 52u, Ba = 137u)

(1) $2NH_4NO_3(s) \rightarrow 2N_2(g) + 4H_2O(g) + O_2(g)$

(2) $Ba(N_3)_2(s) \rightarrow Ba(s) + 3N_2(g)$

(3) $(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(g)$

(4) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

PB0128

52. Lithium aluminium hydride reacts with silicon tetrachloride to form :-**[JEE (Main) ONLINE 2018]**

(1) LiCl, $AlCl_3$ and SiH_4 (2) LiCl, AlH_3 and SiH_4

(3) LiH, $AlCl_3$ and $SiCl_2$ (4) LiH, AlH_3 and SiH_4

PB0129

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53. Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y' Compounds 'X' and 'Y' and the oxidation state of Xe are respectively : **[JEE (Main) ONLINE 2018]**

(1) XeO_2F_2(+6) and XeO_2(+4)
(2) $XeOF_4$(+6) and XeO_2F_2(+6)
(3) $XeOF_4$(+6) and XeO_3(+6)
(4) XeO_2(+4) and XeO_3(+6)

PB0130

54. Which of the following is a lewis acid ? **[JEE (Main) ONLINE 2018]**

(1) NaH (2) NF_3 (3) PH_3 (4) $B(CH_3)_3$

PB0131

55. Iodine reacts with concentrated HNO_3 to yield Y along with other products. The oxidation state of iodine in Y, is :- **[JEE (Main) ONLINE 2019]**

(1) 5 (2) 3 (3) 1 (4) 7

PB0132

56. Among the following reactions of hydrogen with halogens, the one that requires a catalyst is : **[JEE (Main) ONLINE 2019]**

(1) $H_2 + I_2 \rightarrow 2HI$
(2) $H_2 + F_2 \rightarrow 2HF$
(3) $H_2 + Cl_2 \rightarrow 2HCI$
(4) $H_2 + Br_2 \rightarrow 2HBr$

PB0133

57. Correct statements among a to d regarding silicones are : **[JEE (Main) ONLINE 2019]**

(a) They are polymers with hydrophobic character
(b) They are biocompatible.
(c) In general, they have high thermal stability and low dielectric strenth.
(d) Usually, they are resistant to oxidation and used as greases.

(1) (a), (b) and (c) only
(2) (a), and (b) only
(3) (a), (b), (c) and (d)
(4) (a), (b) and (d) only

PB0134

58. Diborane (B_2H_6) reacts independently with O_2 and H_2O to produce, respectively **[JEE (Main) ONLINE 2019]**

(1) HBO_2 and H_3BO_3
(2) H_3BO_3 and B_2O_3
(3) B_2O_3 and H_3BO_3
(4) B_2O_3 and $[BH_4]^-$

PB0135

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59. The amorphous form of silica is : **[JEE (Main) ONLINE 2019]**

(1) quartz (2) kieselguhr (3) cristobalite (4) tridymite

PB0136

60. The correct statements among I to III regarding group 13 element oxides are,

(I) Boron trioxide is acidic. **[JEE (Main) ONLINE 2019]**

(II) Oxides of aluminium and gallium are amphoteric.

(III) Oxides of indium and thalliumare basic.

(1) (I), (II) and (III) (2) (II) and (III) only

(3) (I) and (III) only (4) (I) and (II) only

PB0137

61. The redox reaction among the following is : **[JEE (Main) ONLINE 2020]**

(1) Combination of dinitrogen with dioxygen at 2000 K

(2) Formation of ozone from atmosphereic oxygen in the presence of sunlight

(3) Reaction of H_2SO_4 with NaOH

(4) Reaction of $[Co(H_2O)_6]Cl_3$ with $AgNO_3$

PB0138

62. In the following reactions products(A) and (B), respectively , are : **[JEE (Main) ONLINE 2020]**

$NaOH + Cl_2 \rightarrow$ (A) + side products

(hot and conc.)

$Ca(OH)_2 + Cl_2 \rightarrow$ (B) + side products

(dry)

(1) $NaClO_3$ and $Ca(OCl)_2$ (2) NaOCl and $Ca(ClO_3)_2$

(3) $NaClO_3$ and $Ca(ClO_3)_2$ (4) NaOCl and $Ca(OCl)_2$

PB0139

63. Chlorine reacts with hot and concentrated NaOH and produces compounds (X) and (Y). Compound (X) gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is _____. **[JEE (Main) ONLINE 2020]**

PB0140

64. White Phosphorus on reaction with concentrated NaOH solution in an inert atmosphere of CO_2 gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound (Y) is : **[JEE (Main) ONLINE 2020]**

(1) 4 (2) 1 (3) 2 (4) 3

PB0141

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65. When gypsum is heated to 393 K, it forms : **[JEE (Main) ONLINE 2020]**

(1) Dead burnt plaster

(2) Anhydrous $CaSO_4$

(3) $CaSO_4.5H_2O$

(4) $CaSO_4.0.5H_2O$

PB0142

66. The reaction of $H_3N_3B_3Cl_3$ (A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$. Compounds (B) and (C) respectively, are:

(1) Boron nitride and MeBr **[JEE (Main) ONLINE 2020]**

(2) Borazine and MeMgBr

(3) Borazine and MeBr

(4) Diborane and MeMgBr

PB0143

67. 'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is : **[JEE (Main) ONLINE 2020]**

(1) Carbon tetrachloride (2) Mercury

(3) Silicon carbide (4) Zinc sulphide

PB0144

68. The compound that cannot act both as oxidising and reducing agent is :

(1) H_2O_2 (2) H_2SO_3 **[JEE (Main) ONLINE 2020]**

(3) HNO_2 (4) H_3PO_4

PB0145

69. Match List-I with List-II : **[J-Main 2021]**

List-I Industrial process	**List-II** Application
(a) Haber's process	(i) HNO_3 synthesis
(b) Ostwald's process	(ii) Aluminium extract ion
(c) Contact process	(iii) NH_3 synthesis
(d) Hall-Heroult process	(iv) H_2SO_4 synthesis

Choose the correct answer from the options given below :

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

PB0219

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70. A group 15 element, which is a metal and forms a hydride with strongest reducing power among group 15 hydrides. The element is : **[J-Main 2021]**

(1) Sb (2) P (3) As (4) Bi

PB0220

71. The number of ionisable hydrogens present in the product obtained from a reaction of phosphorus trichloride and phosphonic acid is: **[J-Main 2021]**

(1) 3 (2) 0 (3) 2 (4) 1

PB0221

72. Chemical nature of the nitrogen oxide compound obtained from a reaction of concentrated nitric acid and P_4O_{10} (in 4 : 1 ratio) is : **[J-Main 2021]**

(1) acidic (2) basic (3) amphoteric (4) neutral

PB0222

73. Given below are the statements about diborane **[J-Main 2021]**

(a) Diborane is prepared by the oxidation of $NaBH_4$ with I_2

(b) Each boron atom is in sp^2 hybridized state

(c) Diborane has one bridged 3 centre-2-electron bond

(d) Diborane is a planar molecule

The option with correct statement(s) is -

(1) (c) and (d) only (2) (a) only (3) (c) only (4) (a) and (b) only

PB0223

74. Match List - I with List - II : **[J-Main 2021]**

List - I (compound)	**List - II (effect/affected species)**
(a) Carbon monoxide	(i) Carcinogenic
(b) Sulphur dioxide	(ii) Metabolized by pyrus plants
(c) Polychlorinated biphenyls	(iii) Haemoglobin
(d) Oxides of Nitrogen	(iv) Stiffness of flower buds

Choose the correct answer from the options given below :

(1) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii) (2) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(3) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv) (4) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

PB0224

75. The incorrect statement is: **[J-Main 2021]**

(1) Cl_2 is more reactive than ClF.

(2) F_2 is more reactive than ClF.

(3) On hydrolysis ClF froms HOCl and HF.

(4) F_2 is a stronger oxidizing agent than Cl_2 in aqueous solution

PB0225

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76. Consider the following reactions : **[J-Main 2022]**

$PCl_3 + H_2O \rightarrow A + HCl$

$A + H_2O \rightarrow B + HCl$

number of ionisable protons present in the product B_____. **PB0239**

77. The gas produced by treating an aqueous solution of ammonium chloride with sodium nitrite is

[J-Main 2022]

(1) NH_3 (2) N_2 (3) N_2O (4) Cl_2

PB0240

78. The interhalogen compound formed from the reaction of bromine with excess of fluorine is a :

[J-Main 2022]

(1) hypohalite (2) halate (3) perhalate (4) halite

PB0241

79. Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "X" with 6-equivalents of "Y". "X" and "Y", respectively are : **[J-Main 2022]**

(1) $B(OH)_3$ and NH_3

(2) B_2H_6 and NH_3

(3) B_2H_6 and HN_3

(4) NH_3 and B_2O_3 **PB0242**

80. Dinitrogen and dioxygen. the main constituents of air do not react with each other in atmosphere to form oxides of nitrogen because **[J-Main 2022]**

(1) N_2 is unreactive in the condition of atmosphere.

(2) Oxides of nitrogen are unstable.

(3) Reaction between them can occur in the presence of a catalyst.

(4) The reaction is endothermic and require very high temperature. **PB0243**

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EXERCISE # J-ADVANCED

(IIT JEE ASKED QUESTIONS)

Fill in the blanks

1. Give reason : **[IIT- 2000]**

Why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorus is a tetra atomic molecule.

PB0146

2. Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction.

[IIT- 2000]

PB0147

3. Compounds X on reduction with $LiAlH_4$ gives a hydride Y containing 21.72% hydrogen alongwith other products. The compound Y reacts with air explosively resulting in boron trioxide. Identify X and Y. Give balanced reactions involved in the formation of Y and its reaction with air Draw the structure of Y. **[IIT- 2001]**

PB0148

4. Starting from $SiCl_4$, prepare the following in steps not exceeding the number given in parenthesis (reactions only) **[IIT- 2001]**

(i) Silicon (1)

(ii) Linear silicon containing methyl group only (4)

(iii) Na_2SiO_3 (3)

PB0149

5. Write the balanced chemical equation for developing photographic films. **[IIT- 2001]**

PB0150

6. Identify (X) in the following synthetic scheme and write their structures. **[IIT- 2001]**

$Ba\overset{*}{C}O_3 + H_2SO_4 \longrightarrow X$ (gas) (C denotes C^{14})

PB0151

7. Write the balanced equations for the reactions of the following compounds with water **[2002]**

(i) Al_4Cl_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_3

PB0152

8. Write the balanced equations for the reactions of the following compounds with water: **[IIT- 2002]**

(i) Al_4C_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_4

PB0153

9. Identify the following:

$$\underset{(aq)}{Na_2CO_3} \xrightarrow{SO_2} A \xrightarrow{Na_2CO_3} B \xrightarrow[\Delta]{\text{elemental S}} C \xrightarrow{I_2} D$$

[IIT- 2003]

Also mention the oxidation state of S in all the compounds.

PB0154

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10. Arrange the following oxides in the increasing order of Bronsted basicity. **[IIT- 2004]**
Cl_2O_7, BaO,SO_3, CO_2, B_2O_3

PB0155

11. The number of P—O—P bonds in cyclic tetrametaphosphoric acid is – **[IIT-2000]**
(A) Zero (B) Two (C) Three (D) Four

PB0156

12. The correct order of acidic strength is – **[IIT- 2000]**
(A) $Cl_2O_7 > SO_2 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$
(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

PB0157

13. Amongst H_2O, H_2S, H_2Se and H_2Te, the one with the highest boiling point is – **[IIT- 2000]**
(A) H_2O because of hydrogen bonding (B) H_2Te because of higher molecular weight
(C) H_2S because of hydrogen bonding (D) H_2Se because of lower molecular weight.

PB0158

14. Ammonia can be dried by – **[IIT- 2000]**
(A) Conc. H_2SO_4 (B) P_4O_{10} (C) CaO (D) Anhydrous $CaCl_2$

PB0159

15. Which of the following are hydrolysed – **[REE 2000]**
(A) NCl_3 (B) BCl_3 (C) CCl_4 (D) $SiCl_4$

PB0160

16. The set with correct order of acidity is – **[IIT- 2001]**
(A) $HClO < HClO_2 < HClO_3 < HClO_4$ (B) $HClO_4 < HClO_3 < HClO_2 < HClO$
(C) $HClO < HClO_4 < HClO_3 < HClO_2$ (D) $HClO_4 < HClO_2 < HClO_3 < HClO$

PB0161

17. The reaction, $3ClO^-(aq) \longrightarrow ClO_3^-(aq) + 2Cl^-(aq)$ is an example of – **[IIT- 2001]**
(A) Oxidation reaction (B) reduction reaction
(C) Disproportionation reaction (D) Decomposition reaction

PB0162

18. The number of S–S bonds in sulphur trioxide trimer, (S_3O_9) is – **[IIT- 2001]**
(A) Three (B) Two (C) One (D) Zero

PB0163

19. **Statement-I :** Between $SiCl_4$ and CCl_4, only $SiCl_4$ reacts with water **[IIT- 2001]**
Because :
Statement-II : $SiCl_4$ is ionic and CCl_4 is covalent
(A) If both assertion and reason are correct and reason is the corect explanation of the assertion
(B) If both assertion and reason are correct, but reason is not the correct explanation of the assertion
(C) If assertion is correct, but reason is incorrect
(D) If assertion is incorrect, but reason is correct.

PB0164

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20. Polyphosphates are used as water softening agents because they – **[IIT- 2002]**

(A) Form soluble complexes with anionic species

(B) Precipitate anionic species

(C) Form soluble complexes with cationic species

(D) Precipitate cationic species

PB0165

21. Identify the correct order of solubility of Na_2S, CuS, and ZnS in aqueous medium – **[IIT- 2002]**

(A) CuS > ZnS > Na_2S (B) ZnS > Na_2S > CuS

(C) Na_2S > CuS > ZnS (D) Na_2S > ZnS > CuS

PB0166

22. Identify, the correct order of acidic strength of CO_2, CuO, CaO, H_2O – **[IIT- 2002]**

(A) CaO < CuO < H_2O < CO_2 (B) H_2O < CuO < CaO < CO_2

(C) CaO < H_2O < CuO < CO_2 (D) H_2O < CO_2 < CaO < CuO

PB0167

23. H_3BO_3 is – **[IIT- 2002, 3]**

(A) Monobasic acid and weak Lewis acid (B) Monobasic and weak Bronsted acid

(C) Monobasic and strong Lewis acid (D) Tribasic and weak Bronsted acid

PB0168

24. When I^- is oxidised by MnO_4^- in alkaline medium, I^- converts into – **[IIT- 2003]**

(A) IO_3^- (B) I_2 (C) IO_4^- (D) IO^-

PB0169

25. **Column-I (Change)** **Column-II (Given change is done by)** **[IIT- 2003]**

(A) $Bi^{3+} \longrightarrow (BiO)^+$ (p) Heat

(B) $[AlO_2]^- \longrightarrow Al(OH)_3$ (q) Hydrolysis

(C) $SiO_4^{4-} \longrightarrow Si_2O_7^{6-}$ (r) Acidification

(D) $(B_4O_7^{2-}) \longrightarrow [B(OH)_3]$ (s) Dilution by water

PB0170

26. $(Me)_2 SiCl_2$ on hydrolysis will produce – **[IIT- 2003]**

(A) $(Me)_2 Si(OH)_2$ (B) $(Me)_2 Si = O$

(C) $[—(Me)_2 Si—O—]_n$ (D) $Me_2 SiCl(OH)$

PB0171

27. Which is the most thermodynamically stable allotropic form of phosphorus? **[IIT- 2004]**

(A) Red (B) White (C) Black (D) Yellow

PB0172

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28. When PbO_2 reacts with conc. HNO_3 the gas evolved may be : **[2005]**

(A) NO_2 (B) O_2 (C) N_2 (D) N_2O

PB0173

29. Which of the following is not oxidised by O_3? **[IIT- 2005]**

(A) KI (B) $FeSO_4$ (C) $KMnO_4$ (D) K_2MnO_4

PB0174

30. Which blue-liquid is obtained on reacting equimolar amounts of two gases at – 30ºC ? **[IIT- 2005]**

(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5

PB0175

31. $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$ how can this reaction is made to proceed in forward direction ? **[IIT- 2006]**

(A) Addition of cis 1, 2 diol (B) Addition of borax

(C) Addition of trans 1, 2 diol (D) Addition of Na_2HPO_4

PB0176

32. Among the following, the paramagnetic compound is – **[IIT- 2007]**

(A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2

PB0177

33. **Statement-I** : Boron always forms covalent bond **[2007]**

Because :

Statement-II : The small size of B^{3+} favours formation of covalent bond.

(A) Statement-I is True, Statement-II is True, Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is Ture, Statement-II is not a correct explanation for Statement-II

(C) Statement-I is True, Statement-II is False

(D) Statement-I is False, Statement-II is True

PB0178

34. **Statement-I** : In water, orthoboric acid behaves as a weak monobasic acid. **[2007]**

Statement-II : In water, orthoboric acid acts as a proton donor.

(A) Statement-I is True, Statement-II is True, Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True, Statement-II is not a correct explanation for Statement-II

(C) Statement-I is True, Statement-II is False (D) Statement-I is False, Statement-II is True

PB0179

Comprehension # 1 (Q. 35 to 37)

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling point of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions. The direct reaction of xenon with fluorine leads to a series of compounds with oxidation number + 2, + 4 and + 6. XeF_4 reacts violently with water to give XeO_3. The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell. **[IIT- 2007]**

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35. Argon is used in arc welding because of its –

(A) Low reactivity with metal (B) Ability to lower the melting point of metal

(C) Flammability (D) High calorific value

PB0180

36. The structure of XeO_3 is –

(A) Linear (B) Planar (C) Pyramidal (D) T-shaped

PB0181

37. XeF_4 and XeF_6 are expected to be –

(A) Oxidising agent (B) Reducing agent (C) Unreactive (D) Strongly basic

PB0182

Comprehension # 2 (Q.38 to 40)

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3. Phosphine is a flammable gas and is prepared from white phosphorous.

[IIT- 2008]

38. Among the following, the correct statement is :-

(A) Phosphates have no biological significance in humans

(B) Between nitrates and phosphates, phosphates are less abundant in earth's crust

(C) Between nitrates and phosphates, nitrates are less abundant in earth's crust

(D) Oxidation of nitrates is possible in soil

PB0183

39. Among the following, the correct statement is :-

(A) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

(B) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

(C) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

(D) Between NH_3 and PH_3. PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

PB0184

40. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a :-

(A) dimerization reaction (B) disproportionation reaction

(C) condensation reaction (D) precipitation reaction

PB0185

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41. The reaction of P_4 with X leads selectively to P_4O_6. The X is **[JEE 2009]**

(A) Dry O_2 (B) A mixture of O_2 and N_2

(C) Moist O_2 (D) O_2 in the presence of aqueous NaOH

PB0186

42. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively **[JEE 2012]**

(A) redox reaction ; -3 and –5 (B) redox reaction ; +3 and +5

(C) disproportionation reaction ; -3 and +1 (D) disproportionation reaction ; -3 and +3

PB0187

43. Bleaching powder contains a salt of an oxoacid as one of its components. The anhydride of that oxoacid is :

(A) Cl_2O (B) Cl_2O_7 (C) ClO_2 (D) Cl_2O_6 **[JEE 2012]**

PB0188

44. With respect to graphite and diamond, which of the statement(s) given below is (are) correct ? **[JEE 2012]**

(A) Graphite is harder than diamond.
(B) Graphite has higher electrical conductivity than diamond.
(C) Graphite has higher thermal conductivity than diamond.
(D) Graphite has higher C–C bond order than diamond.

PB0189

45. Concentrated nitric acid, upon long standing, turns yellow-brown due to the formation of -

(A) NO (B) NO_2 (C) N_2O (D) N_2O_4 **[JEE 2013]**

PB0190

46. The correct statement(s) about O_3 is(are) **[JEE 2013]**

(A) O–O bond lengths are equal (B) Thermal decomposition of O_3 is endothermic

(C) O_3 is diamagnetic in nature (D) O_3 has a bent structure

PB0191

Comprehension # 3 (Q. 47 and 48)

The reaction of Cl_2 gas with cold dilute and hot concentrated NaOH in water give sodium salt of two (different) oxoacids of chlorine P and Q respectively. The Cl_2 gas reacts with SO_2 gas , in presence of charcoal to give a product R. R reacts with white phosphorous to give a compound S. On hydrolysis, S gives as oxoacid of phosphorous T.

47. R, S and T , respectively are - **[JEE 2013]**

(A) SO_2Cl_2, PCl_5 and H_3PO_4 (B) SO_2Cl_2, PCl_3 and H_3PO_3

(C) $SOCl_2$, PCl_3 and H_3PO_2 (D) $SOCl_2$, PCl_3 and H_3PO_3

PB0192

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48. P and Q, respectively, are the sodium salts of -

(A) Hypochlorus and chloric acid (B) Hypochlorus and chlorus acid

(C) Chloric and perchloric acids (D) Chloric and hypochlorus acids

PB0193

49. The unbalanced chemical reactions given in List-I show missing reagent or condition (?) which are provided in List-II. Match List-I with List-II and select the correct answer using the code given below the lists :A **[JEE 2013]**

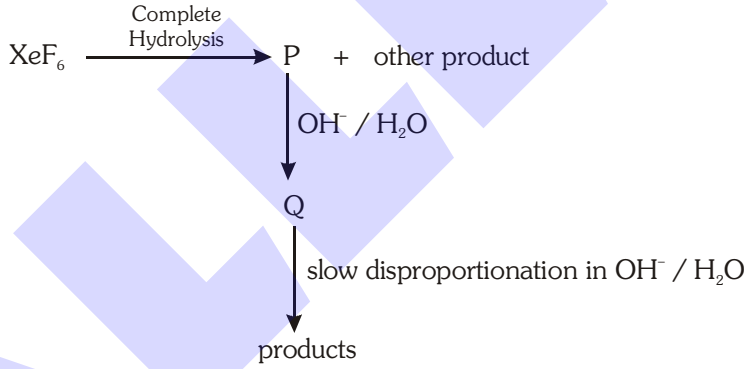
	List-I		**List-II**
(P)	$PbO_2 + H_2SO_4 \xrightarrow{?} PbSO_4 + O_2$ + other product	(1)	NO
(Q)	$Na_2S_2O_3 + H_2O \xrightarrow{?} NaHSO_4$ + other product	(2)	I_2
(R)	$N_2H_4 \xrightarrow{?} \overset{\circ}{N}_2$ + other product	(3)	Warm
(S)	$XeF_2 \xrightarrow{?} Xe$ + other product	(4)	Cl_2

Codes :

	P	Q	R	S		P	Q	R	S
(A)	4	2	3	1	(B)	3	2	1	4
(C)	1	4	2	3	(D)	3	4	2	1

PB0194

50. Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is **[JEE Adv. 2014]**



(A) 0 (B) 1 (C) 2 (D) 3

PB0195

51. The product formed in the reaction of $SOCl_2$ with white phosphorous is **[JEE Adv. 2014]**

(A) PCl_3 (B) SO_2Cl_2 (C) SCl_2 (D) $POCl_3$

PB0196

52. The **CORRECT** statements(s) for orthoboric acid is / are - **[JEE Adv. 2014]**

(A) It behaves as a weak acid in water due to self ionization

(B) Acidity of its aqueous solution increases upon addition of ethylene glycol

(C) It has a three dimensional structure due to hydrogen bonding.

(D) It is a weak electrolyte in water

PB0197

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53. The correct statement(s) regarding, (i) $HClO$, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is(are)
(A) The number of Cl=O bonds in (ii) and (iii) together is two **[JEE Adv. 2015]**
(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
(C) The hybridization of Cl in (iv) is sp^3
(D) Amongst (i) to (iv), the strongest acid is (i)

PB0198

54. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2. The **TRUE**, statement (s) regarding this adsorption is (are) **[JEE Adv. 2015]**
(A) O_2 is physisorbed (B) heat is released
(C) occupancy of π^*_{2p} of O_2 is increased (D) bond length of O_2 is increased

PB0199

55. Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are **[JEE (Adv.) 2015]**
(A) CH_3SiCl_3 and $Si(CH_3)_4$ (B) $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$
(C) $(CH_3)_2SiCl_2$ and CH_3SiCl_3 (D) $SiCl_4$ and $(CH_3)_3SiCl$

PB0200

56. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is - **[JEE (Adv.) 2015]**

PB0201

57. The increasing order of atomic radii of the following group 13 elements is : **[JEE Adv. 2016]**
(A) Al < Ga < In < Tl (B) Ga < Al < In < Tl
(C) Al < In < Ga < Tl (D) Al < Ga < Tl < In

PB0202

58. The crystalline form of borax has **[JEE Adv. 2016]**
(A) Tetranuclear $[B_4O_5(OH)_4]^{2-}$ unit
(B) All boron atoms in the same plane
(C) Equal number of sp^2 and sp^3 hybridized boron atoms
(D) One terminal hydroxide per boron atom

PB0203

59. The nitrogen containing compound produced in the reaction of HNO_3 with P_4O_{10}
(A) can also be prepared by reaction of P_4 and HNO_3 **[JEE Adv. 2016]**
(B) is diamagnetic
(C) contains one N-N bond
(D) reacts with Na metal producing a brown gas

PB0204

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60. The correct statements(s) about the oxoacids, $HClO_4$ and HClO, is (are) - **[JEE Adv. 2017]**

(A) $HClO_4$ is more acidic than HClO because of the resonance stabilization of its anion

(B) $HClO_4$ is formed in the reaction between Cl_2 and H_2O

(C) The central atom in Both $HClO_4$ and HClO is sp^3 hybridized

(D) The conjugate base of $HClO_4$ is weaker base than H_2O

PB0205

61. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to - **[JEE Adv. 2017]**

(A) the physical state of X_2 at room temperature changes from gas to solid down the group

(B) decrease in HOMO-LUMO gap down the group

(C) decrease in π^*-σ^* gap down the group

(D) decrease in ionization energy down the group

PB0206

62. The order of the oxidation state of the phosphorus atom in H_3PO_2, H_3PO_4, H_3PO_3 and $H_4P_2O_6$ is **[JEE Adv. 2017]**

(A) $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$

(B) $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$

(C) $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$

(D) $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

PB0207

63. The option(s) with only amphoteric oxides is (are): **[JEE Adv. 2017]**

(A) Cr_2O_3, CrO, SnO, PbO

(B) NO, B_2O_3, PbO, SnO_2

(C) Cr_2O_3, BeO, SnO, SnO_2

(D) ZnO, Al_2O_3, PbO, PbO_2

PB0208

64. Among the following, the correct statement(s) is are **[JEE Adv. 2017]**

(A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure

(B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure

(C) BH_3 has the three-centre two-electron bonds in its dimeric structure

(D) The Lewis acidity of BCl_3 is greater than that of $AlCl_3$

PB0209

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Paragraph for Q.65 & 66

Upon heating $KClO_3$ in the presence of catalytic amount of MnO_2 , a gas **W** is formed. Excess amount of **W** reacts with white phosphorus to give **X.** The reaction of **X** with pure HNO_3 gives **Y** and **Z**.

[JEE Adv. 2017]

65. **W** and **X** are, respectively

(A) O_3 and P_4O_6 (B) O_2 and P_4O_{10} (C) O_3 and P_4O_{10} (D) O_2 and P_4O_6

PB0210

66. **Y** and **Z** are , respectively

(A) N_2O_4 and H_3PO_3 (B) N_2O_4 and HPO_3

(C) N_2O_5 and HPO_3 (D) N_2O_3 and H_3PO_4

PB0210

67. The compound(s) which generate(s) N_2 gas upon thermal decomposition below 300°C is (are)

(IIT JEE 2018)

(A) NH_4NO_3 (B) $(NH_4)_2Cr_2O_7$ (C) $Ba(N_3)_2$ (D) Mg_3N_2

PB0211

68. The total number of compounds having at least one bridging oxo group among the molecules given below is_____. **[JEE Adv. 2018]**

N_2O_3, N_2O_5, P_4O_6, P_4O_7, $H_4P_2O_5$, $H_5P_3O_{10}$, $H_2S_2O_3$, $H_2S_2O_5$

PB0212

69. A tin chloride **Q** undergoes the following reactions (not balanced) **[JEE Adv. 2019]**

Q + $Cl^- \rightarrow$ **X**

Q + $Me_3N \rightarrow$ **Y**

Q + $CuCl_2 \rightarrow$ **Z** + CuCl

X is a monoanion having pyramidal geometry. Both **Y** and **Z** are neutral compounds. Choose the correct option(s).

(1) The central atoms in **X** is sp^3 hybridized

(2) The oxidation state of the central atom in **Z** is +2

(3) The central atom in **Z** has one lone pair of electrons

(4) There is a coordinate bond in **Y**

PB0213

70. At 143 K. the reaction of XeF_4 with O_2F_2 produces a xenon compound **Y**. The total number of lone pair(s) of electrons present on the whole molecule of **Y** is _____ **[JEE Adv. 2019]**

PB0214

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71. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc.HNO_3 to a compound with the highest oxidation state of sulphur is ___ **[JEE Adv. 2019]**

(Given data : Molar mass of water = 18 g mol^{-1}) **PB0215**

72. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE Adv. 2019]**

(1) $BeCl_2$, CO_2, BCl_3, $CHCl_3$ (2) SO_2, C_6H_5Cl, H_2Se, BrF_5

(3) BF_3, O_3, SF_6, XeF_6 (4) NO_2, NH_3, $POCl_3$, CH_3Cl **PB0216**

73. Among B_2H_6, $B_3N_3H_6$, N_2O, N_2O_4, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is ———— **[JEE Adv. 2019]**

PB0217

74. Choose the correct statement(s) among the following. **[JEE Adv. 2020]**

(A) $SnCl_2.2H_2O$ is a reducing agent.

(B) SnO_2 reacts with KOH to form $K_2[Sn(OH)_6]$.

(C) A solution of $PbCl_2$ in HCl contains Pb^{2+} and Cl^- ions.

(D) The reaction of Pb_3O_4 with hot dilute nitric acid to give PbO_2 is a redox reaction.

PB0229

75. Which of the following liberates O_2 upon hydrolysis? **[JEE Adv. 2020]**

(A) Pb_3O_4 (B) KO_2 (C) Na_2O_2 (D) Li_2O_2

PB0230

76. With respect to hypochlorite, chlorate and perchlorate ions, **[JEE Adv. 2020]**

choose the correct statement(s).

(A) The hypochlorite ion is the strongest conjugate base.

(B) The molecular shape of only chlorate ion is influenced by the lone pair of electrons of Cl.

(C) The hypochlorite and chlorate ions disproportionate to give rise to identical set of ions.

(D) The hypochlorite ion oxidizes the sulfite ion.

PB0231

77. The correct statement(s) related to oxoacids of phosphorous is(are) **[JEE Adv. 2021]**

(A) Upon heating, H_3PO_3 undergoes disproportionation reaction to produce H_3PO_4 and PH_3.

(B) While H_3PO_3 can act as reducing agent, H_3PO_4 cannot.

(C) H_3PO_3 is a monobasic acid.

(D) The H atom of P-H bond in H_3PO_3 is not ionizable in water.

PB0232

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78. Ozonolysis of ClO_2 produces an oxide of chlorine. The average oxidation state of chlorine in this oxide is ___. **[JEE Adv. 2021]**

PB0233

79. The compound(s) which react(s) with NH_3 to give boron nitride (BN) is(are) **[JEE Adv. 2022]**

(A) B (B) B_2H_6 (C) B_2O_3 (D) HBF_4

PB0244

80. The reaction of $HClO_3$ with HCl gives a paramagnetic gas, which upon reaction with O_3 produces

(A) Cl_2O (B) ClO_2 (C) Cl_2O_6 (D) Cl_2O_7 **[JEE Adv. 2022]**

PB0245

81. The reaction of Xe and O_2F_2 gives a Xe compound **P**. The number of moles of HF produced by the complete hydrolysis of 1 mol of **P** is _____. **[JEE Adv. 2022]**

PB0246

82. Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives a gas **Q**. The amount of $CuSO_4$ (in g) required to completely consume the gas **Q** is _____.

[Given : Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63] **[JEE Adv. 2022]**

PB0247

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ANSWER KEY

EXERCISE # I

(ONLY ONE OPTION IS CORRECT)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	B	A	B	B	D	C	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	B	B	A	B	A	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	D	A	A	B	B	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	D	A	B	C	C	B	C	B	D
Que.	41	42	43	44	45	46	47			
Ans.	A	A	B	C	A	C	A			

EXERCISE # II

(ONE OR MORE THAN ONE OPTION MAY BE CORRECT)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, B	A, B, C	A, C, D	A, B, C	A, B, D	A, B	A, B, C	A, B, C	A,B,C,D	A, C, D
Que.	11	12	13	14	15	16	17			
Ans.	A,B,C,D	C, D	C, D	A,B,C,D	D	B, C, D	A,B,C,D			

EXERCISE # III

PARAGRAPH & MATRIX MATCH

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	D	A	C	D	A	B	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	C	B	B	C	A	D	C
Que.	21	22	23	24						
Ans.	BCD	B	A	B						

EXERCISE # J-MAINS

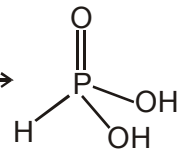
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	2	3	4	2	2	2	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	3	1	4	3	2	3	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	2	1	3	4	2	2	3	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	1	1	3	4	4	2	2	3	2
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	2	2	4	4	2	1	3	3	3	3
Que.	51	52	53	54						
Ans.	4	1	2	4						

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Que.	55	56	57	58	59	60	61	62	63	64
Ans.	1	1	3	3	2	1	1	1	1.66 to 1.67	2
Que.	65	66	67	68	69	70	71	72	73	74
Ans.	4	2	1	4	3	4	3	1	2	1
Que.	75									
Ans.	1									

76. (2)

Sol. $PCl_3+H_2O \xrightarrow[\text{hydrolysis}]{\text{Partial}} PCl_2(OH)\text{ (or) }PCl(OH)_2 + HCl$

$PCl_2(OH)\text{ (or) }PCl(OH)_2$ (A) $\xrightarrow{\text{water}}$  (B) $+ HCl$

no. of ionisable protons in B = 2

77. (2)

Sol. $NH_4Cl + NaNO_2 \rightarrow NH_4NO_2 + NaCl$

$\downarrow$

$N_2 + 2H_2O$

78. (2)

Sol. $Br_2 + 5\underset{\text{(excess)}}{F_2} \longrightarrow 2\overset{+5}{Br}F_5 \xrightarrow{H_2O} HBrO_3$ (Forms bromate)

79. (2)

Sol. $3B_2H_6 + 6NH_3 \xrightarrow{\Delta} 2B_3N_3H_6 + 12\ H_2$

80. (4)

Sol. $N_2 + O_2 \underset{}{\overset{\text{(1483-2000 K)}}{\rightleftharpoons}} 2NO$ (Endothermic and feasible at high temperature)

EXERCISE # J- ADVANCED

Fill in the blanks

1. $P \equiv P$

$3p_\pi$-$3p_\pi$ bond is not effective due to large size of P.

2. $2I^-\text{ (aqueous)} + Cl_2 \longrightarrow I_2 + 2Cl^-\text{(aqueous)}$

(i) $2I^-\text{ (aqueous)} \longrightarrow I_2\text{ (s)} + 2e^-$

(ii) $Cl_2\text{ (g)} + 2e^- \longrightarrow 2Cl^-\text{ (aq)}$

Thus, I^- is oxidised into I_2 by Cl_2 due to higher oxidised potential of Cl_2 than I_2

3. X : BCl_3

Y : B_2H_6

$4BCl_3 + 3LiAlH_4 \longrightarrow 3AlCl_3 + 3LiCl + 2B_2H_6$

X

$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$ (exothermic)

Y

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4. (i) $3\,SiCl_4 + 4Al \longrightarrow 3Si + 4AlCl_3$ (in one step)

(ii) $SiCl_4 + 2Mg \longrightarrow 2MgCl_2 + Si$

$Si + Cu \longrightarrow Si - Cu$

$2CH_3Cl + Si - Cu \longrightarrow (CH_3)_2SiCl_2 + Cu$

$(CH_3)_2SiCl_2 + 2H_2O \longrightarrow (CH_3)_2Si(OH)_2 + 2HCl$

$$2(CH_3)_2Si(OH)_2 \xrightarrow[-H_2O]{\Delta} H{-}O{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{Si}}{-}O{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{Si}}{-}OH$$

(iii) $SiCl_4 + 4H_2O \longrightarrow Si(OH)_4 + 4HCl$

$Si(OH)_4 \longrightarrow SiO_2 + 2H_2O$

$SiO_2 + Na_2CO_3 \xrightarrow[1673K]{\Delta} Na_2SiO_3 + CO_2$

5. Unreacted AgBr is removed by hypo ($Na_2S_2O_3$)

$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

6. $^{14}CO_2$

7. (i) $Al_4Cl_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4\uparrow$

(ii) $CaNCN + 3H_2O \longrightarrow CaCO_3\downarrow + 2NH_3$

(iii) $4BF_3 + 3H_2O \longrightarrow \underset{\text{boric acid}}{H_3BO_3} + \underset{\text{fluoboric acid}}{3HBF_4}$

(iv) $NCl_3 + 3H_2O \longrightarrow NH_3 + \underset{\text{hypochlorous acid}}{3HOCl}$

(v) $2XeF_4 + 3H_2O \longrightarrow Xe + \underset{\text{xenon trioxide}}{XeO_3} + F_2 + 6HCl$

8. (i) $Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$

(ii) $CaNCN + 3H_2O \longrightarrow CaCO_3 + 2NH_3$

(iii) $BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3HF$

$3HF + 3BF_3 \longrightarrow 3HBF_4$

$4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3HBF_4$

---

9. Oxidation state

(A) : $NaHSO_3$ + 4

(B) : Na_2SO_3 + 4

(C) : $Na_2S_2O_3$ + 2

(D) : $Na_2S_4O_6$ + 2.5

10. $Cl_2O_7 < SO_3 < CO_2 < B_2O_3 < BaO$

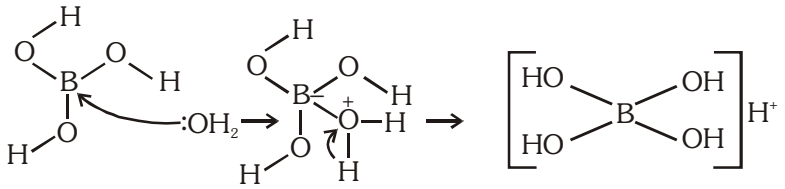
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	A	C	ABD	A	C	D	C	C
Que.	21	22	23	24	25				26	27
Ans.	D	A	A	A	A→Q,S ; B→R,S ; C→P ; D→Q,R				C	C
Que.	28	29	30	31	32					
Ans.	B	C	B	A	D					

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33. A

Boron always forms covalent bond because boron requires very high energy of form B^{3+} and again B^{3+} due to its very small size having high polarising power thus cause greater polarisation and eventually significant covalent characteristics-Fajans rule.

34. C



Que.	35	36	37	38	39	40	41	42	43	44
Ans.	A	C	A or B	C	C	B	B	C	A	B,D
Que.	45	46	47	48	49	50	51	52	53	54
Ans.	B	A,C,D	A	A	D	C	A	B,D	B,C	B,C,D
Que.	55	56								
Ans.	B	6								

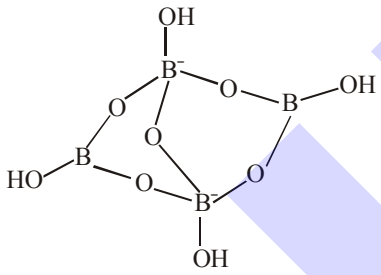
57. **(B)**

Sol. The order of radius of 13th group elements is Ga < Al < In < Tl.

Reason ⇒ Due to poor shielding effect of d-orbital, radius of Ga is smallar than Al.

58. **(A,C,D)**

Sol.



(A) Having $[B_4O_5(OH)_4]^{2-}$ tetranuclear (boron) unit

(B) All boron atoms not in same plane

(C) Two boron are sp^2 hybridised and two boron are sp^3 hybridised

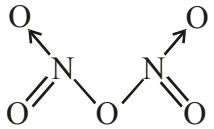
(D) One terminal hydroxide per boron atom is present.

59. **(B,D)**

Sol. $P_4O_{10} + 4HNO_3 \xrightarrow{\text{dehydration of } HNO_3} \underset{\text{(required product)}}{4(HPO_3)} + 2N_2O_5$

(A) $P_4 + 20\,HNO_3 \rightarrow 4H_3PO_4 + 20NO_2\uparrow + 4H_2O$

(B) N_2O_5 is diamagnetic in nature

(C) $N_2O_5 \rightarrow$ 

N_2O_5 contains one N–O–N bond not N–N bond.

(D) $Na + N_2O_5 \rightarrow NaNO_3 + \underset{\text{(Brown gas)}}{NO_2\uparrow}$

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Que.	60	61	62	63	64	65	66	67	68	69
Ans.	ACD	BC	A	CD	ACD	B	C	BC	5 or 6	1,4
Que.	70	71	72	73						
Ans.	19.00	288.00	2,4	4.00						

74. (A,B or A,B,C)

Sol. (A) $SnCl_2.2H_2O$ is a reducing agent since Sn^{2+} tends to convert into Sn^{4+}.

(B) $\underset{\text{(Amphoteric)}}{SnO_2} + 2KOH_{(aq.)} + 2H_2O \longrightarrow \underset{\text{(Base)}}{K_2[Sn(OH)_6]}$

(C) First group cations (Pb^{2+}) form insoluble chloride with HCl that is $PbCl_2$ however it is slightly soluble in water and therefore Pb^{+2} ion is never completely precipitated on adding hydrochloric acid in test sample of Pb^{2+}, rest of the Pb^{2+} ions are quantitatively precipitated with H_2S in acidic medium.
So that we can say that filtrate of first group contain solution of $PbCl_2$ in HCl which contains Pb^{2+} and Cl^-.

However in the presence of conc. HCl or excess HCl it can produce $H_2[PbCl_4]$

So, we can conclude A, B or A,B,C should be answers.

(D) $\underset{\substack{(2PbO.PbO_2)\\\text{(mixture of oxides)}}}{Pb_3O_4} + 4HNO_3 \longrightarrow PbO_2(\downarrow) + 2Pb(NO_3)_2 + 2H_2O$

It is not a redox reaction.

75. (B)

Sol. (A) Pb_3O_4 is insoluble in water or do not react with water.
(B) $2KO_2 + 2H_2O \rightarrow 2KOH + H_2O_2 + O_{2(g)}\uparrow$
(C) $Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$
(D) $Li_2O_2 + 2H_2O \rightarrow 2LiOH + H_2O_2$

76. (A,B,D)

Sol. Hypochlorite ion : $ClO^{\ominus}$

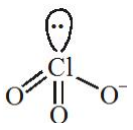
Chlorate ion : $ClO_3^{\ominus}$

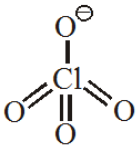
Per chlorate ion : $ClO_4^{\ominus}$

(A) Acidic order : $\overset{+1}{H}ClO < \overset{+5}{H}ClO_3 < \overset{+7}{H}ClO_4$

Conjugate base order : $ClO^- > ClO_3^- > ClO_4^-$

(B) Hypochlorite ion ($ClO^{\ominus}$) : $[\ddot{\underset{\cdot\cdot}{Cl}} - \ddot{\underset{\cdot\cdot}{O}}]^{\ominus}$ Linear shape

Chlorate ion :  Trigonal pyramidal shape

Perchlorate ion :  Perfect tetrahedral shape due to resonance

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In chlorate ion bond angle changes due to presence of lone pair on chlorine atom. While hypochlorite ion is linear and perchlorate ion is tetrahedral and there is no effect of lone pair on hypochlorite ion.

(C) Disproportionation reaction of

(i) hypochlorite ion : $3ClO^{\ominus} \rightarrow 2Cl^{-} + ClO_3^{\ominus}$

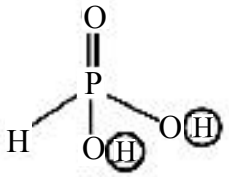
(ii) Chlorate ion : $4ClO_3^{\ominus} \rightarrow 3ClO_4^{\ominus} + Cl^{\ominus}$

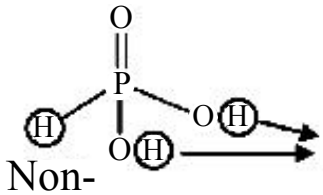
(D) $ClO^{-} + SO_3^{2-} \rightarrow SO_4^{2-} + Cl^{\ominus}$

77. (A,B,D)

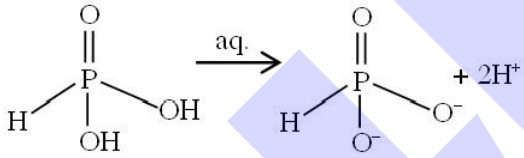
Sol. (A) $4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$ (correct)

(B) H_3PO_4 has "P" in its highest oxidation state, hence cannot act as a reducing agent (correct).

(C)  Dibasic and (incorrect)
Two OH group present in H_3PO_4

(D)  Ionizable (Correct)
Non-ionizable

The hydrogen which is directly attached to phosphorous does not ionized in water.



78. (6)

Sol. $2ClO_2 + 2O_3 \rightarrow Cl_2O_6 + 2O_2$

Cl_2O_6

$2x + 6(-2) = 0$

$x = +6$

Average oxidation state of Cl in Cl_2O_6 is 6.

79. (B,C) or (A,B,C)

Sol. (A) $2B + 2NH_3 \rightarrow 2BN + 3H_2$

Boron produced BN with ammonia but **Boron is element not compound**. So that this option not involve in answer.

(B) $3B_2H_6 + 6NH_3 \rightarrow 3[BH_2(NH_3)_2]^{+}[BH_4^{-}] \xrightarrow{T = 200°C} 2B_3N_3H_6 + 12H_2$

$B_3N_3H_6 \xrightarrow{T > 200°C} (BN)_x$

(C) $B_2O_3(\ell) + 2NH_3 \xrightarrow{1200°C} 2BN_{(s)} + 3H_2O_{(g)}$

(D) $HBF_4 + NH_3 \rightarrow NH_4[BF_4]$

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80. (C)

Sol. $HClO_3 + HCl \rightarrow \underset{\text{(Paramagnetic)}}{ClO_2} + \frac{1}{2}Cl_2 + H_2O$

$2ClO_2 + 2O_3 \rightarrow Cl_2O_6 + 2O_2$

81. (2 or 4 or 6)

Sol. $Xe + 2O_2F_2 \rightarrow XeF_4 + 2O_2$

$3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + \frac{3}{2}O_2 + 12HF$

$\therefore$ One mole of XeF_4 gives 4 moles of HF on hydrolysis.

82. (2.38 / 2.39)

Sol. Mole of $P_4 = \frac{1.24}{31 \times 4} = 0.01$

$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

0.01 mole 0.01 mole

$2PH_3 + 3CuSO_4 \rightarrow Cu_3P_2 + 3H_2SO_4$

$0.01 \quad \frac{3}{2} \times 0.01$

$= \frac{0.03}{2}$ moles

$W_{CuSO_4} = \frac{0.03}{2} \times 159 = 2.385$ gm

Ans. = 2.38 or 2.39

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